

Karsten Lowe

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EDUCATION

The University of California, San Diego

B.S in Data Science, San Diego, California

Expected Graduation, June 2028

- **Related Coursework:** Principles of Data Science, Programming and Data Structures for Data Science, Statistical Analysis, Linear Algebra, Calculus and Analytic Geometry.

EXPERIENCE

AWS- Solutions Architect

December 2025 - Present

- **Deployed** containerized applications using **Docker** with Amazon **ECS** and **ECR**.
- Utilized AWS database services including **RDS**, **Aurora**, **DynamoDB**, **S3**, **Neptune**, and **Timestream**, selecting appropriate **SQL** and **NoSQL** solutions based on workload requirements.
- Implemented **data pipelines**, analytics, and **ML workflows** using **Kinesis** (Streams & Firehose), **SQS**, **Lambda**, **Athena**, **Redshift**, **QuickSight**, and **SageMaker**.

Data Science- Self-directed Learner

August 2025 - Present

- Applied **Python** and **Pandas** to query, clean, merge, and normalize datasets, ensuring high-quality data for **exploratory data analysis (EDA)**, **modeling**, **reporting**, and **visualization**.
- Utilized **scikit-learn** for **data preprocessing**, **pipeline construction**, **cross-validation**, **model evaluation**, and **metric selection**.

University of Texas at Dallas Business Analytics- Scholar

July 2024 - August 2024

- Conducted demographic-based sales analysis of the 99Bikes dataset, using **SQL** to clean, filter, and validate data for high data quality, and built interactive **Tableau** dashboards to visualize key sales trends.
- **Presented** data-driven insights and recommendations to a **graduate-level** Business Analytics audience, demonstrating the ability to translate analytical findings into **actionable** business value.

Math Tutor- Individual

July 2025 - present

- **Provided individualized tutoring (4th grade-Calculus & SAT Math)** for 8-10 hours weekly, using customized worksheets and targeted instruction. **Most students improved by one to two letter grades**.
- Marketed tutoring services through infographics placed in local businesses to **expand** the client base.
- Provide **online tutoring**, sustaining academic support remotely while maintaining **full-time studies**.

PROJECTS

NFL Big Data Bowl - Team Lead & Developer

November- December 2025

- **Developed** a **fully deployable** predictive "best play" interface for the **NFL**, enabling **coaches** and **players** to make higher-impact, real-time **decisions** by optimizing play selection through analysis of team attributes, defensive schemes, and field position.
- Engineered a physics-based positional metric in **Python (Pandas)** with kinematic and directional data to model **optimal** receiver decisions by calculating proximity-based interactions with defensive coverage.
- Transformed user-selected 5-20 yd intervals into advanced route-performance insights by applying **linear regression** to estimate position-based scoring likelihoods, using **bootstrapped** confidence intervals for uncertainty quantification, and performing **permutation tests** to evaluate statistical significance between potential routes.

Awards & Honors

- Top 15 **WIC UCSD** Beginner Programming Competition November 2025
- Recognized **UTD** Business Analytics Presenter August 2024

